



New York University
Computer Science Department
Introduction to Software Development
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Homework 4 *Practicing control statements and Strings.*

Deadline: See NYUclasses for the deadline, 5% off per day after the deadline (3 days maximum).

Please read the guidelines carefully to avoid receiving a zero on the HW:

- Please use the following naming convention “ExerciseX.java”, where X is the number of the exercise.
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercise.
- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly
- **Do not use eclipse or any IDE, only text editors and command line**

Exercise 1 (10pts) Repeat Kiosk problem in Assignment 3 with the following modifications

- Use methods. Each operation is represented by a method
- Use do – while loop to make sure the kiosk is always ready for further transactions – not just one time run–
- If option “4” is entered the program exist.
- Starting balance is \$100.00

Exercise 2 (10pts) Write a Java program that contains a method that takes a number (from the user as an input) and prints it in reverse order. That is, if the user enters 123, the program should print 321.

(you are NOT allowed to use arrays)

Exercise 3 (10pts) Write program in java that accepts user’s input and validates if the input (could be a word or more) is a palindrome. A palindrome is a word that reads the same forwards

as it does backwards. For this exercise we will consider a palindrome a palindrome without punctuation.

Examples of palindromes: My gym, or Top spot , race car

Exercise 4 (10 point) Write a Java program that converts a decimal number to binary based 2. For instance, 45 in binary is 101101. The decimal number is an input from the user.

Exercise 5 (15 pts) Write a java program that contain two overloaded methods that accepts two numbers or two characters representing a range example (11, 37) or (c, w) inputted by the user. The method generates a random integer within the given range and returns it, or a random char within the given range of chars and returns it. The method needs to be called in the main and main prints the return of the method.

Exercise 6 (20 points) Implement a Java program that prints a table of characters that are equivalent to the Ascii codes from 65 to 126 (126 included) (You will need to use int and cast it to char) Your program should be print five characters per line.

Read about Ascii codes:

<http://www.ascii-code.com/>

<http://www.computerhope.com/jargon/a/ascii.htm>

Exercise 7 (25pts) Write a program that prompts the user to enter two strings and displays the largest common prefix of the two strings (ignoring the letters case). Here are some sample runs:

e.g. Enter the first string: Welcome to Java

Enter the second string: Welcome to programming

The common prefix is: Welcome to

e.g. Enter the first string: Atlanta Enter the second string: Miami

Output: Miami and Atlanta have no common prefix